GATE

CS & IT

Digital Logic

DPP: 1

Boolean Theorems & Logical Gates

Q1 $(A+B)(A+C)(A+\overline{C})$ is equivalent to
(A) A + BC
(B) $A+B\overline{C}$
(C) 0
(D) A

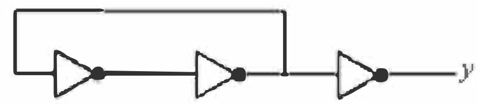
Q2 A logical function is given as:
$$f(A,B,C)=B\overline{C}[A+A\overline{C}D+\overline{B}CD+\overline{A}B\overline{C}+\overline{A}\overline{B}\overline{C}]$$
is equivalent to
(A) $A\overline{B}CD$
(B) $B\overline{C}$
(C) $A\overline{B}+B\overline{C}+CD$
(D) $AB\overline{C}D$

Q3 If we have 4-variables in a logical function, then number of non-dual logical functions possible______.

Q4 A logical function
$f(A, B, C) = (A + B)(\overline{B}+C)(A+C)$, then
$\overline{f}$ will be equal to
(A) $AB+\overline{B}C$
(B) $\overline{A}\overline{B}+B\overline{C}$
(C) $\overline{A}\overline{B}+\overline{AC}$
(D) $AB+AC$

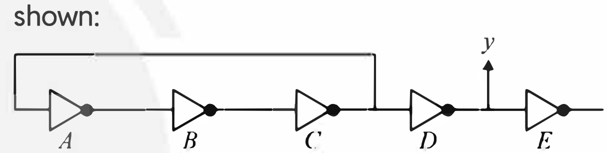
Q5 Which of the following statement is true?
(A) Dual function f^D is always equals to f.
(B) NAND is self dual in nature.
(C) NOT is self dual in nature.
(D) Number of self dual function with 3-variables is *8*.

Q6 A digital circuit is designed as shown:



This circuit works as
(A) Astable multivibrator as odd number of Not gates are used.
(B) Bistable multivibrator
(C) Monostable multivibrator
(D) None of these

Q7 An astable multivibrator circuit is designed as shown:



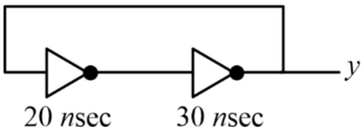
NOT gates A, B & C are of logic family TTL with delay t_d = 20/3 nsec of each gate and gates D & E are of logic family ECL with delay t'_d =10/3 nsec of each gate, then frequency of the waveform at output y is ______ MHz.

Q8 Logical function $f(A, B, C, D)$ =
$AB+\overline{A}CD+\overline{B}CD$ is equivalent to
(A) $AB+\overline{B}C$
(B) AB + CD
(C) $\overline{A}C+\overline{B}C$
(D) $AB+\overline{B}C$

Q9 A logical function is given as:
f(A, B, C) = $\overline{A}\overline{B}+\overline{A}BC+\overline{A}B\overline{C}$
then which of the following statement is true?
(A) f(A, B, C) = $\overline{A}\overline{B}+B\overline{C}$
(B) $f(A,\ B,\ C)=\overline{A}+\overline{C}$
(C) $f(A,\ B,\ C)$ is a self dual function.

(D) None of the above

Q10 A logical circuit is designed as shown:



Output y is

(A) A bistable multivibrator's output

(B) A square waveform with $f = 10$ MHz

(C) A square waveform with $f = 25$ MHz

(D) None of the above.

Q11 Which of the following is true?

(A) $\overline{\overline{A}B + A\overline{B}} = \left(\overline{A} + \overline{B}\right)\left(A + B\right)$

(B) $\overline{AB\,\overline{CD}} = \overline{A} + \overline{B} + \overline{C} + \overline{D}$

(C) $\overline{\overline{A\overline{B}}.C} = \left(A + \overline{C}\right)\left(\overline{B} + \overline{C}\right)$

(D) None of these

Q12 Which of the following is true?

(A) We can use '1' as enable input for OR gate

(B) We can use '0' as enable input for AND gate

(C) '0' as well as '1' can be used as enable input for XNOR gate

(D) None of the these

Q13 Which of the following relation is true?

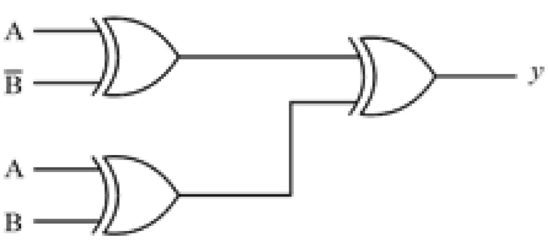
(A) $A \oplus \overline{B} = \overline{A} \odot B$

(B) $\overline{A \oplus \overline{B}} = A \odot B$

(C) $\overline{\overline{A} \odot \overline{B}} = A \oplus B$

(D) $\overline{\overline{A} \oplus \overline{B}} = A \oplus B$

Q14 A logical circuit is as given below:



Output y will be

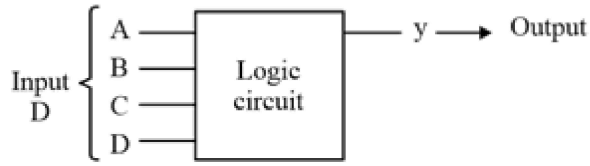
(A) $\overline{A} + B$

(B) $\overline{A} + \overline{B}$

(C) $A\overline{B}$

(D) A + B

Q15 A logic circuit has 4-input & 1-output line as shown:



Output y is '1' wherever no. of zeroes on input side are odd, then output y can be expressed as:

(A) $A \odot B \odot C \odot D$

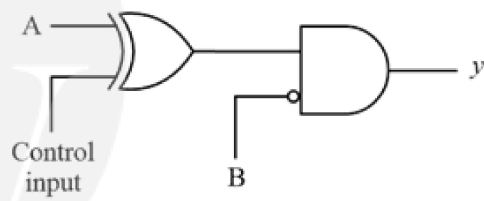
(B) $\overline{A \odot B \odot C} \odot D$

(C) $\overline{A \oplus B} \oplus C \oplus \overline{D}$

(D) None of these

Q16 A logic circuit is as given below:

Which of the following is true?



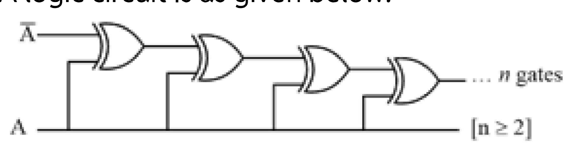
(A) Output y is $\overline{A}B$ if control input = 0

(B) Output y is $\overline{A + B}$ if control input = 1

(C) Output y is $\overline{A \cdot B}$ if control input = 0

(D) Output y is $\overline{\overline{A} \cdot B}$ if control input = 1

Q17 A logic circuit is as given below:



Which of the following is true?

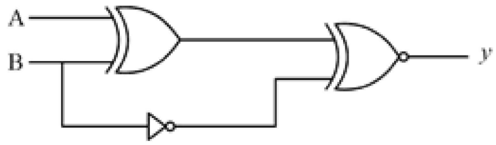
(A) Output is $\overline{A}$ if n is even

(B) Output is A if n is even

(C) Output is $\overline{A}$ if n is odd

(D) Output is A if n is odd

Q18 A logical circuit is as given below:



Output y is

(A) A (B) $\overline{B}$

(C) $\overline{A}$ (D) B

Q19 A logical expression is given as:

$f(A, B, C, D) = \overline{A} + AB[ABC + \overline{B}C + AB\overline{C} + C\overline{D}]$

then minimum number of 2-input NAND gate require to implement above logic function will be _____ .

Q20 A logical expression is given as:

$f(A, B, C) = (\overline{A} + B)(A + \overline{B})$,

minimum number of 2-input NAND gate require to implement above logical function is .

Q21 A logical expression is given as:

$f(A, B, C) = \overline{A} + ABC$, then minimum number of 2-input NAND gate require to implement above logical function is ______ .

Q22 A logical function is given as:

$f(A, B) = A \oplus A\overline{B}$, If we implement this logical function using 2-input NAND gate then, minimum number of NAND gate require is

Answer Key

Q1	(D)
Q2	(B)
Q3	65280~65280
Q4	(B)
Q5	(C)
Q6	(B)
Q7	25~25
Q8	(B)
Q9	(C)
Q10	(A)
Q11	(C)
Q12	(C)
Q13	(C)
Q14	(B)
Q15	(B, C)
Q16	(B)
Q17	(A)
Q18	(A)
Q19	2~2
Q20	5~5
Q21	2~2
Q22	2~2